



الاختبار النصفى لمقرر التعلم العميق المتقدم (AI4204)

Midterm Exam – Advance Deep Learning (AI4203)

الفصل الدراسي	تاريخ الاختبار	مدة الاختبار
472	الأربعاء 1447/10/20 الموافق 2026/04/08	ساعة

تعليمات الاختبار:

- عدد أسئلة الاختبار ١٢ سؤالاً.
- درجة الاختبار ٣٠ درجة.
- تأكد من كتابة اسمك ورقمك الجامعي ورقم الشعبة في ورقة الإجابة .
- غير مسموح باستخدام أي مذكرات أو مراجع أثناء الاختبار (Closed book exam).
- في حال احتجت إلى مساعدة أو استفسار، ارفع يدك وانتظر المراقب للرد علي ك.
- احرص على متابعة وقت الاختبار لكي تتمكن من الإجابة على جميع الأسئلة
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Exam Instructions

- The exam consists of 12 questions.
- The total exam score is 30 marks.
- Make sure to clearly write your name, university ID number, and section number on the answer sheet.
- No notes or reference materials are allowed during the exam (closed-book exam).
- At the end of the exam, submit both the question paper and the answer sheet to the invigilator.
- If you need assistance or have a question, raise your hand and wait for the invigilator to assist you.
- Please manage your time carefully to ensure you are able to answer all required questions.

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1. What are the three main parts that make up the Transformer architecture?

Tokenizer $\times \rightarrow$ جزء من مرحلة معالجة البيانات
Encoder و ليس جزء من ال
Decoder المقترضا ذكر ال
Transformer Architecture
Attention

2. What is the primary function of "Positional Encoding" in Transformers?

To provide the transformer with the order of the words or position in the sequence

ممتاز
الوقت، المقارنة، الترتيب

3. In calculating self-attention, three variables taken from database terminology are used. What are they?

Q.K.V

Query
Key
Value

4. In the self-attention mechanism, what does the Query (Q) represent?

- A) The stored information retrieved as a result of the attention process
- ☒ B) The current word that is "asking" which other words are most relevant to it
- C) A label used to index and match words in the attention lookup table
- D) The scaling factor applied before the SoftMax to stabilize gradients

5. The SoftMax function in the self-attention equation converts similarity scores into probabilities that sum to 1, producing attention weights for each word.

- ☒ A) True
- B) False



6. What is the key difference between Self-Attention and Masked Self-Attention regarding access to other words?

- A) Self-Attention can only look at previous words, while Masked Self-Attention can look at all words
- ☒ B) Self-Attention allows every word to attend to all other words, while Masked Self-Attention restricts each word to attend only to previous words
- C) Both allow full access to all words, but Masked Self-Attention applies dropout for regularization
- D) Masked Self-Attention attends to future words only, while Self-Attention attends to past words only

7. Why do we transpose the Key matrix (K) when multiplying it with the Query

- A) To normalize the values and prevent large numbers
- ☒ B) To make the matrix dimensions compatible for multiplication
- C) To reduce the size of the matrix for faster computation
- D) To apply positional encoding to the input

8. What is the title of the seminal 2017 paper that first introduced transformer architecture, and what specific task was the model originally designed to perform?

✓ Attention Is All you need
✗ Classification
↑ Machine translation

9. In Encoder-Decoder Attention, where do Q, K, V come from?

Q comes From Decoder

K and V from Encoder

Q: Query
K: Key
V: Value

10. Mention two practical uses for the "Context Aware Embeddings" produced by Encoder-Only

Transformers?

Sentiment analysis

Named Entity Recognition

Text classification

مطلوبين 2
و ذكرين 3
أنف شار



11. Explain what Word Embedding and Positional Encoding are, and how they are combined to form the input of a Transformer model.

Word embedding gives you the meaning of the words
Positional Encoding tells you the place of the word
or the order of the words. In the sequence, they are
combined so that the model can understand and know
the order of the words cause otherwise it process
it all at the same time without knowing the order or
the meaning of it. It could say: the man bit the dog
Instead of: the dog bit the man

12. Compare Encoder-Only and Decoder-Only Transformers. Explain the type of attention used in each, their primary goals, and why ChatGPT is considered a Generative Model.

Aspect	Encoder-Only	Decoder-Only
Attention	Self-attention	Masked Self-attention
Goal	to get the context aware embeddings	to generate the next word
Example tasks	Sentiment analysis Named Entity Recognition	Generative models (ChatGPT)

Why ChatGPT is generative:

because it uses decoder-only model to generate the next word based on the previous words without knowing the next words

الله يفتح عليك
اجاباتك ممتازة

~ Good Luck ~



Questions Overview (Marks Distribution)

Q#	Question Type	Marks
Q1	Short Answer	2
Q2	Short Answer	2
Q3	Short Answer	2
Q4	MCQ	2
Q5	True/False	2
Q6	MCQ	2
Q7	MCQ	2
Q8	Short Answer	3
Q9	Short Answer	2
Q10	Short Answer	2
Q11	Essay	4
Q12	Essay	5
Total		30